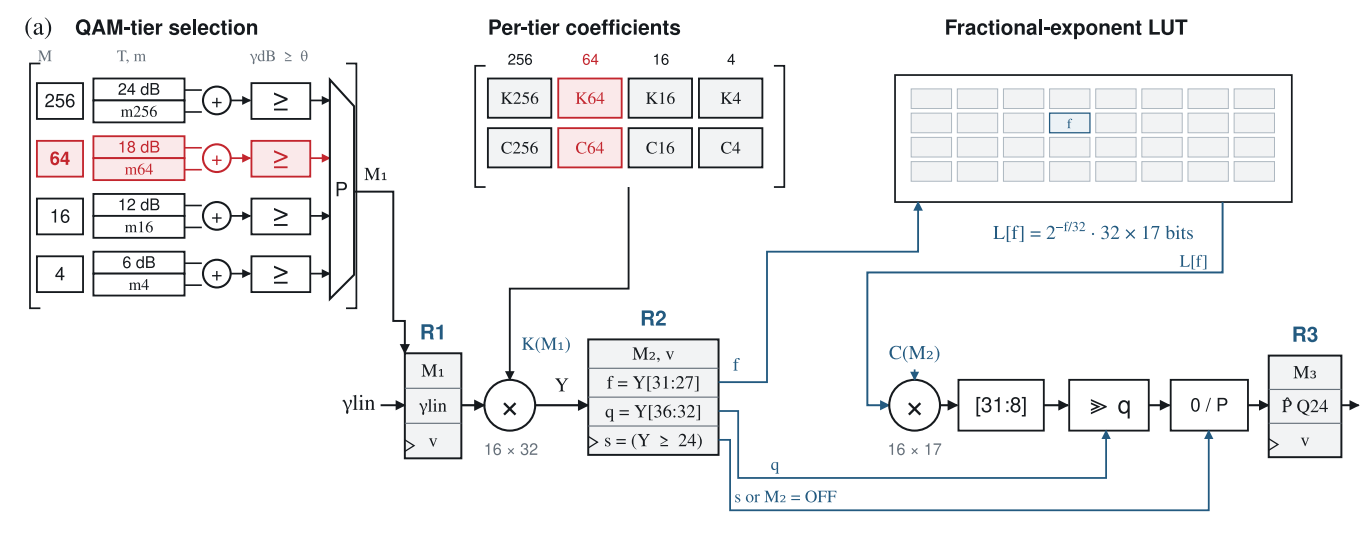

(a) QAM-tier selection
M
T, m
γdB ≥ θ
256
24 dB
m256
64
18 dB
m64
16
12 dB
m16
4
6 dB
m4
P
M₁
Per-tier coefficients
256
64
16
4
K256
K64
K16
K4
C256
C64
C16
C4
Fractional-exponent LUT
f
L[f] = 2^−f/32 · 32 × 17 bits
L[f]
R1
M₁
γlin
v
γlin
K(M₁)
16 × 32
Y
R2
M₂, v
f = Y[31:27]
q = Y[36:32]
s = (Y ≥ 24)
f
C(M₂)
16 × 17
[31:8]
≫ q
0 / P
R3
M₃
P̂ Q24
v
q
s or M₂ = OFF